



機電整合與虛擬實境實驗 Mechatronics and XR Lab 2026

Lab 3

從 Arduino 走向電腦端程式

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Today's Four Tasks

A software-focused lab: firmware, desktop apps, and wireless comms



1 Basic Task 3-1 Timer Interrupt

TimerOne library — fires on a fixed period, no external signal needed



2 Advanced Task 3-1 Timer Interrupt vs Blocking Delay

Compare timer-interrupt responsiveness against delay()



3 Advanced Task 3-2 LED Control with Serial Communication

UART serial communication between a host PC app (C# console) and Arduino



4 Advanced Task 3-3 LED Control with Wireless Communication

Using an HC05 module to implement LED control through wireless communication

Three Things to Remember Before You Start



**Timer1 units are
microseconds**

500ms means entering 500000,
not 500



**AI-generated code is
allowed**

But you must be able to explain
the architecture during your
demo



TX / RX logic levels differ

Arduino runs at 5V, HC-05 at
3.3V — a voltage divider on RXD
is recommended



Task Check & Submission Rules

From Lab 3, you only need to demonstrate and submit Advanced Task files



What's new from Lab 3

- **Only Advanced Task** files need to be demonstrated and submitted from now on
- Basic Tasks are still strongly recommended — they build the skills Advanced Tasks depend on
- Call the TA over and briefly explain your sketch and circuit yourself
- The TA may ask follow-up questions



For **Advanced Task** submission

- Upload the **sketch file** and a **demo video** to GitHub — after your task check
- Create a "Lab 3" folder. Suggested naming: "Task3-1.ino", "Task3-1.mp4"
- Confirm with the lecturer or TA that your files uploaded successfully before you leave the classroom

Basic Task 3-1 Timer Interrupt with Arduino

Adding an external library isn't always smooth — try to troubleshoot it yourself if you hit issues!

🎯 Demo goal

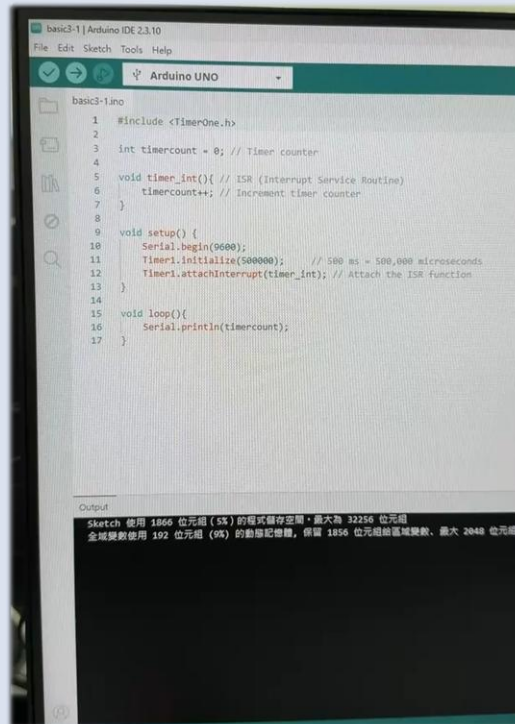
Declare `int timercounter = 0`, then use the `TimerOne` library to trigger a timer interrupt every 500ms, incrementing it each time.



Hint

Two of the methods to download the external library “**TimerOne**” : from the library manager in Arduino IDE or through the website:

<https://code.google.com/archive/p/arduino-timerone/downloads>



```
#include <TimerOne.h>

int timercount = 0; // Timer counter

void timer_int(){ // ISR (Interrupt Service Routine)
    _____; // Increment timer counter
}

void setup() {
    Serial.begin(9600);
    Timer1.initialize(____);
    Timer1.attachInterrupt(____);
}

void loop(){
    Serial.println(timercount);
}
```

Advanced Task 3-1 Timer Interrupt vs Blocking Delay

Refer to Basic Task 1-2 and build two button + LED pairs



Button A + LED A

Timer Interrupt (TimerOne)

- TimerOne library sets up a periodic interrupt every 50ms
- Read Button A's state and update LED A inside the ISR



Button B + LED B

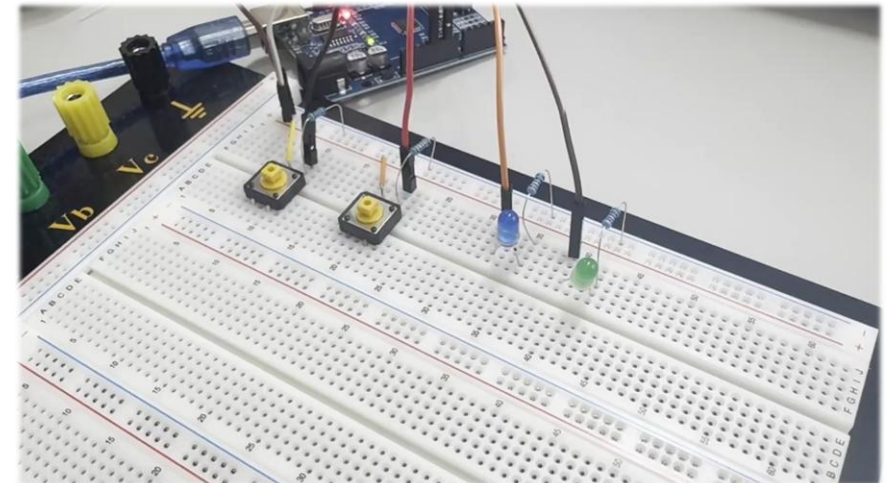
Blocking Delay

- Read Button B's state and update LED B every loop
- Insert a 1 second at the end of “ `void loop()` ” function.



Demonstrate & explain to the TA

- What is a timer interrupt? What are the differences between the two methods in this task?
- Which method do you prefer to use for developing a robotic system?

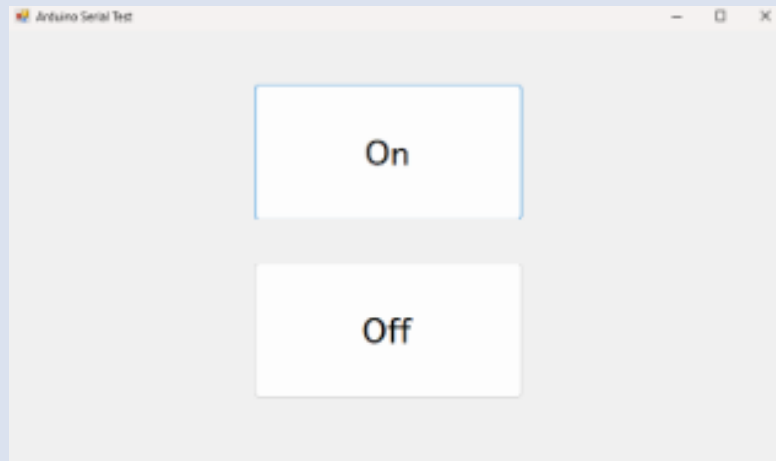


Advanced Task 3-2 LED Control with Serial Communication

Two-way UART communication between a C# host app and Arduino

<> Host PC App (C#)

- Design a GUI to control the LED
- When a button is pressed, send the corresponding value through the serial port



Arduino Firmware (.ino)

- Initialize Serial.begin(9600)
- Continuously listen for incoming serial strings
- Turn the LED on / off according to the data received from the PC app



Advanced Task 3-2 Development Hints



Break it into subtasks

This task is more complex than previous ones, but valuable for real project development. Split it into smaller pieces to complete it.



Choose the right project template

Use Visual Studio to build the PC host app. Investigate which project template fits this task yourself.



AI tools are allowed

You may use AI to help generate the sketch and source code — but you must understand it and be ready to explain it to the TA.



You don't need to master C#

You don't need to understand every function or command. Focus on understanding the code's structure and how it works.

Advanced Task 3-3 HC-05 Wireless LED Control

🎯 Demo goal

- Using an HC05 module to implement LED control through wireless communication
- Complete the previous task again, but this time with Bluetooth communication.
- The Arduino receives commands from the host PC application using an HC-05 module.
- The key to completing this task is configuring the HC05 Bluetooth module.
- You may also need to revise the Arduino sketch and C# code accordingly.



Stuck? Troubleshooting Checklist



LED blinks erratically

`Timer1.initialize()` unit mistake —
microseconds entered as milliseconds



C# can't connect to the port

Make sure the Arduino IDE's Serial Monitor
isn't holding the port open



No response after pairing

Confirm the AT-mode baud rate matches the
normal-operation baud rate



Upload fails

The serial port is held by another program
— close any other windows using it

Lab 3 Complete! Your Core Hardware Toolkit Is in Place

Next → Final Project Integration

Q & A